

React Basics

Build UIs with Components and JSX

■ LEARNING OUTCOME

By the end of this module you will create a React project with Vite, understand JSX syntax, write functional components, render lists, handle conditional rendering, and understand the Virtual DOM.

■ Skills You Will Gain

- Create a React + Vite project and understand file structure
- Write JSX and understand its rules
- Create functional components
- Render lists with `.map()` and key prop
- Use conditional rendering patterns
- Understand Virtual DOM and reconciliation

■ React is used by Facebook, Instagram, Airbnb, Netflix, and thousands of startups. It is the most in-demand frontend skill globally and the fastest path to a frontend job.

SECTION 1

Setup and JSX

```
# Create React + Vite project npm create vite@latest my-app -- --template react cd my-app && npm install && npm run dev // JSX Rules: // 1. Return ONE root element (wrap in <> Fragment) // 2. Use className not class // 3. Self-close void elements: <img /> <input /> // 4. Expressions in { } curly braces // 5. Attributes camelCase: onClick, onChange, tabIndex // 6. Inline styles: style={{ color:'red', fontSize:16 }} function WelcomeBanner() { const user = { name:'Alice', role:'admin' }; const isLoggedIn = true; const items = ['React','TypeScript','Vite']; return ( <> {isLoggedIn ? <img src='/avatar.jpg' alt='avatar' /> : <div>?</div> } <h1>Welcome, {user.name}!</h1> <p style={{ color:'purple', fontWeight:'bold' }}>Role: {user.role.toUpperCase()}</p> <ul> {items.map((item, i) => <li key={i}>{item}</li>)} </ul> </> ); }
```

SECTION 2

Functional Components and Lists

```
// Basic component function Greeting({ name, age = 18 }) { return <div><h2>Hello, {name}!</h2><age && <p>Age: {age}</p></div>; } // Arrow function component const ProductCard = ({ title, price, inStock, onAddToCart }) => ( <article className='product-card'> <h3>{title}</h3> <span>Rs. {price.toLocaleString()}</span> <span className={inStock ? 'badge-green' : 'badge-red'}> {inStock ? 'In Stock' : 'Out of Stock'} </span> <button onClick={onAddToCart} disabled={!inStock}>Add to Cart</button> </article> ); // Conditional rendering patterns const StatusBadge = ({ status }) => { const config = { delivered: { color:'green', label:'Delivered' }, shipped: { color:'blue', label:'Shipped' }, processing: { color:'orange', label:'Processing' }, }; const { color, label } = config[status] || { color:'gray', label:status }; return <span className={`badge badge-${color}`}>{label}</span>; }; // List with empty state const TodoList = ({ todos }) => ( <ul> {todos.length === 0 ? <li>No todos yet!</li> : todos.map(todo => ( <li key={todo.id} className={todo.done ? 'completed' : ''}> {todo.text} </li> )) } </ul> ); export default ProductCard; export { Greeting, TodoList };
```

SECTION 3

Virtual DOM

Why React Uses a Virtual DOM

Real DOM manipulation is expensive -- each change can trigger layout recalculation and repaint.\n\nReact keeps a VIRTUAL DOM (lightweight JS object copy of real DOM).\n\nWhen state changes:\n1. React creates a NEW Virtual DOM tree\n2. React DIFFS new vs old virtual DOM (reconciliation)\n3. React applies ONLY the minimal real DOM changes needed\n\nKEY prop helps React identify which list items changed, were added, or removed.\nBad key (index): can't tell which item changed when list reorders.\nGood key (stable ID): React directly knows which item to update.

■ PRACTICE Q & A

Q1. What is JSX and is it required?

A: JSX is a syntax extension that lets you write HTML-like code in JS. It compiles to React.createElement() calls. Not required but universally used for readability.

Q2. Why must list items have a unique key prop?

A: React uses keys to identify changed items between renders. Without keys, React re-renders all items on any change. Use stable unique IDs, not array indices.

Q3. What is a Fragment and when do you use it?

A: A Fragment (<> </>) lets you return multiple elements without adding an extra DOM node. Use when sibling elements would affect CSS layout if wrapped in a div.

Q4. What is the difference between className and class in JSX?

A: class is a reserved JS keyword (for ES6 classes). JSX uses className which compiles to the HTML class attribute.

Q5. What is reconciliation in React?

A: React's process of comparing new Virtual DOM with previous one to determine minimum changes needed. Makes updates $O(n)$ through heuristics about component identity.

■■ React Basics Cheat Sheet	
<code>npm create vite@latest app -- --template react</code>	Create React + Vite project
<code>function Comp({ prop }) { return <jsx/> }</code>	Functional component
<code>className='cls'</code>	CSS class in JSX
<code>onClick={handler}</code>	Event handler camelCase
<code>{ expression }</code>	JavaScript in JSX
<code>{condition && <El />}</code>	Render if condition true
<code>{cond ? <A/> : }</code>	Ternary conditional render
<code>items.map(i => <li key={i.id}>)</code>	List -- use stable ID as key
<code><> children </></code>	Fragment -- no DOM node
<code>export default Comp</code>	Default export
<code>import Comp from './Comp'</code>	Default import
<code>loading='lazy'</code>	Lazy-load images